DANİEL YAMANOGLU

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Scan for portfolio

PROFILE

Control and Automation Engineer specializing in the convergence of classical Control Theory and modern Generative AI architectures. I build intelligent systems that bridge hardware control logic with autonomous AI, leveraging a strong foundation in state estimation and stochastic modeling to drive scalable, data-centric workflows.
Core Focus Areas:
• Architecting robust RAG pipelines and LLM orchestration frameworks.
• Training and optimizing ML models (XGBoost, Neural Networks) for predictive analytics.
• Engineering data ingestion pipelines to transform raw hardware/sensor metrics into actionable AI features.
• Developing high-load, production-ready backends using Python, FastAPI, and Docker.

EXPERIENCES

03/2026 – 06/2026 **Engineering Intern | ENKO**
• Evaluated PLC architectures and mapped sensor data flow across industrial protocols (e.g., Modbus, MQTT).
• Engineered telemetry data ingestion pipelines for predictive maintenance models.

06/2025 – 09/2025 **Engineering Intern | ADATECH**
• Developed an HR analytics dashboard featuring Optimized Team and Risk Analysis modules.
• Built an XGBoost-based predictive model for early detection of turnover risks with 91.7% accuracy.
• Designed data-driven workflows, improving workforce allocation efficiency by 15%.

EDUCATION

2020 – 2026 **Control and Automation Engineering (GPA: 3.01 / 4.00)**
Yıldız Technical University

2016 – 2020 **Fener Rum High School**

SKILLS

• **AI/ML:**
PyTorch, XGBoost, LangChain

• **Languages:**
Python, C/C++, MATLAB

• **Infrastructure:**
ROS, Docker, Kubernetes, CI/CD, FastAPI, Simulink

• **Data:**
ChromaDB, Qdrant, Pinecone, Neo4j, Voyage

LANGUAGES

English	Greek	Turkish	Arabic
C1	Native	Native	A2

PROJECTS

Designed an ML-powered system with XGBoost algorithm to predict employee turnover and visualize team dynamics.
• Integrated SHAP values for model explainability, providing visual dashboards for team dynamics and risk analysis.

Graduation Thesis: EKF-Integrated Fuzzy-CTC Control for 2-DOF Robot Manipulator
• Developed a hybrid architecture combining CTC with Fuzzy logic to dynamically compensate for non-linear uncertainties.

Life OS: Agentic Knowledge System
• Architected a personal knowledge operating system utilizing Neo4j graph databases for complex semantic relationship mapping.

Matchfluence: Influencer Marketing Marketplace
• Engineered the scalable backend infrastructure for a B2B SaaS platform utilizing Supabase.

CERTIFICATES

Control Design Onramp with Simulink   Simulink Onramp MATLAB Onramp

• IEEE - Power in Energy
• IEEE - RLC Days
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